Certified Learning for Lossy Power Grids: Compositional Verification of a Learned Stability Correction

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Abstract

A stability certificate is only worth the set it names, and on a lossy grid that set is expensive to prove, since branch-and-bound cost doubles with every generator, while losses destroy the classical energy function that would otherwise make certification cheap. We certify a learned stability correction on the lossy Kron-reduced New England 39-bus system by composing per-machine certificates, eliminating the speed variables in closed form and proving a 9-D angle condition by sound interval branch-and-bound. Monolithic verification of the same claim needs on the order of 10^{18} leaf boxes at 20-D where the compositional certificate needs about 10^3 operations, a measured $7 \times 10^{14}\times$ reduction. The correction must be a vector term rather than a potential, because the field required to cancel the lossy coupling is 98% rotational and no gradient can supply it, and learning it enlarges the certificate to 24.8% mean coverage of the true region of attraction at 100% soundness against 23.2% for the physics baseline. An independently implemented exact sum-of-squares verifier ties the interval result on every slice, the same procedure transfers to a separately built topology at $+38\%$ relative, and one convex program recovers $+86\%$ relative coverage by fixing the radius and asking only the shape question. Composition buys the radius at polynomial cost, and what remains is the geometry of the certificate rather than the tightness of the verifier.

1 Introduction

Transient stability asks whether a grid returns to synchronous operation after a disturbance, and from how large a set of post-fault states. The classical answer is an energy function whose sublevel sets lie provably inside the region of attraction, and it exists only for lossless networks, because the lossy term $G_{ij}\cos(\delta_{ij})$ is not the gradient of any scalar potential (Chiang et al., 1995). Learning a Lyapunov function replaces the missing energy function with a trained one and certifies it by bound propagation (Chang et al., 2019; Zhang et al., 2018), but its cost grows exponentially in state dimension, so lossy benchmarks of this size have not been certified before. Exact sum-of-squares programming is the other route, and it grows combinatorially in degree and dimension (Papachristodoulou and Prajna, 2002; Henrion and Korda, 2014).

We certify a learned correction on the full ten-generator lossy benchmark by making the certificate compositional, reusing the Kron-reduced IEEE New England 39-bus plant and its closed-loop equilibrium verbatim (Cui and Zhang, 2022; Athay et al., 1979; Chow) and building on compositional certification of networked systems (Zhang et al., 2023) and dissipativity-based composition for power systems (Yang et al., 2025). Four measured claims carry the result: composition pulls the verification cost of one claim from 10^{18} leaf boxes to 10^3 operations; the correction cannot be a scalar potential, because the required field is 98% rotational, and a learned vector cross term enlarges the certified region from 23.2% to 24.8% at 100% soundness; an independently implemented exact verifier lands on the same numbers on every slice, which locates the remaining gap in the certificate rather than in

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the bounding; and because radius and shape are separate questions, asking the shape question alone with the radius fixed recovers $+86\%$ relative coverage from one convex program.

2 Problem formulation and the lossy obstruction

Plant. The IEEE New England 39-bus system is Kron-reduced to its ten generator buses (Chow), giving the swing model

$$M_i\ddot{\delta}_i = P_i(\boldsymbol{\delta}) - d_i\dot{\delta}_i, \qquad P_i(\boldsymbol{\delta}) = P_{m,i} - \textstyle\sum_j \big[F_{ij}\sin\delta_{ij} + G_{ij}\cos\delta_{ij}\big], \tag{1}$$

with $\delta_{ij} = \delta_i - \delta_j$, M_i inertia, d_i droop damping, $P_{m,i}$ mechanical input, F_{ij} the conservative susceptance coupling and G_{ij} the transfer conductance (zero in the lossless limit). Generator 0 supplies the angle reference, leaving a 9-D relative angle vector $\boldsymbol{\phi}$ and a 9-D relative speed vector $\boldsymbol{\omega}$ around the equilibrium $\boldsymbol{\delta}^*$. We estimate the region of attraction by dense Monte Carlo integration on three held-out 2-D angle slices, the pairs $(5,6)$, $(4,7)$ and $(1,9)$, with the remaining active angles pinned at equilibrium ($n = 5{,}041$ points per slice). For a certified set $\mathcal{C}$, coverage is $|\mathcal{C} \cap \mathrm{stable}|/|\mathrm{stable}|$ averaged over the slices, and soundness is 100% minus the share of certified points later found unstable.

Certificate family and speed elimination. Consider

$$V(\boldsymbol{\phi},\boldsymbol{\omega}) = \tfrac{1}{2}\boldsymbol{\omega}^\top M\boldsymbol{\omega} + U(\boldsymbol{\phi}) + \boldsymbol{\Psi}(\boldsymbol{\phi})^\top M\boldsymbol{\omega}, \tag{2}$$

where $M = \mathrm{diag}(M_i)$, U is the classical transient energy potential of the conservative coupling, and $\boldsymbol{\Psi}$ is a vector field we call the cross term. Because (2) is quadratic in $\boldsymbol{\omega}$, differentiating along (1) gives $\dot{V} = \boldsymbol{\omega}^\top Q(\boldsymbol{\phi})\boldsymbol{\omega} + b(\boldsymbol{\phi})^\top\boldsymbol{\omega} + c_0(\boldsymbol{\phi})$ with $Q = -\mathrm{diag}(d) + \Omega\,\mathrm{sym}(MJ)$, $J = \partial\boldsymbol{\Psi}/\partial\boldsymbol{\phi}$, $b_i = -L_i(\boldsymbol{\phi}) - d_i\Psi_i(\boldsymbol{\phi})$, where $L_i = \sum_j G_{ij}(\cos\delta_{ij} - \cos\delta^*_{ij})$ is the lossy residual, and $c_0 = \boldsymbol{\Psi}^\top P$. Where $Q \prec 0$ the maximum over *every* speed is closed form,

$$h(\boldsymbol{\phi}) := \max_{\boldsymbol{\omega}} \dot{V} = \tfrac{1}{4}\, b(\boldsymbol{\phi})^\top\big(-Q(\boldsymbol{\phi})\big)^{-1} b(\boldsymbol{\phi}) + c_0(\boldsymbol{\phi}), \tag{3}$$

so proving $h(\boldsymbol{\phi}) < 0$ over an angle-only box proves $\dot{V} < 0$ for every speed at every angle in it. The speed dimension is removed analytically rather than searched, which is half of why composition is cheap.

Why no scalar correction can work. With $\boldsymbol{\Psi} \equiv 0$ and U the lossless potential, the residual L_i enters (3) squared and positive, so it can only make the certified threshold harder to reach. A scalar residual $\Phi(\boldsymbol{\phi})$ contributes only a gradient to b, while the field required to cancel L has curl $\nabla \times F_{\mathrm{req}} = 2G_{ij}\sin\delta_{ij}$, measured at about 98% of that field's magnitude. A gradient cannot approximate a field that is almost entirely rotational, whatever the capacity or the training procedure, and the ledger below records the consequence, which is that the scalar lever alone scores 9.4%, worse than the physics baseline.

3 Method

The learned vector cross term. Moving the correction into $\boldsymbol{\Psi}$ removes the integrability requirement, because cancelling b becomes the pointwise algebraic assignment $\Psi_i = -L_i/d_i$, a condition on a function's values rather than on its being a gradient, so the rotational part of L is reachable. Each Ψ_i is a shallow one-hidden-layer $\tanh$ network over local port variables, and the monotone activation is load-bearing, because its interval image is exact and its Jacobian closed form, which keeps the sound bounds tight. It starts at the affine physics baseline $\Psi_i = \varepsilon\phi_i$ and trained counterexample-guided against h evaluated through the verifier's own relaxation on refined boxes, because a static coarse cover cannot train this field, since any network pays an over-approximation penalty that on coarse boxes dominates whatever it gains on the exact h, so the optimum is to switch it off. A log-barrier floor on the structural coupling margins keeps them at 100% of the physics field, and constraining them *improved* coverage from 24.3% to 24.8% rather than costing it.

Sound verification at compositional cost. Condition (3) is verified by interval branch-and-bound run through the network. $(-Q)^{-1}$ is never interval-inverted directly, since $-Q = D - E$ is split into diagonal and off-diagonal parts and E is dominated by a per-row Gershgorin bound in Loewner order, giving a sound closed-form upper bound on h per box. A small central cube ($|\phi_i| \le \rho = 0.05$) is excluded from branch-and-bound because $h \to 0$ there by construction, and is covered instead

by a local negative-definite-Hessian argument plus dense sampling, the one sampled ingredient in the study, identical across every certificate reported here. Monolithic branch-and-bound over the 18-D state needs about 1.15×10^{18} leaf boxes while the compositional certificate reaches the same claim in about 1.6×10^{3} operations, a measured $7 \times 10^{14}\times$ reduction (Figure 1d). All reported certificates keep both Lyapunov conditions proved, $h < 0$ by branch-and-bound and $V > 0$ on the Schur complement $V_{\text{eff}} = V - \frac{1}{2}\mathbf{\Psi}^{\top} M \mathbf{\Psi}$ over each proved box.

4 Results

Training uses six slices disjoint from the three evaluation slices, with model selection worst-case over training slices only, so no evaluation slice is touched by training or selection.

Result 1: the learned cross term provably enlarges the certificate. The physics baseline certifies proved radii of $102.0°$, $96.4°$ and $75.1°$ on the three held-out slices, 23.2% mean coverage at 100% soundness. The learned cross term, the only change, enlarges the radius on two of three slices to $106.4°$ and $103.1°$ and holds the third at $72.9°$, for 24.8% mean coverage, 100% sound and zero certified-unstable points (Figure 1a,b). Dense evaluation of the exact unrelaxed condition $h < 0$, rather than any bound, shows that a perfect zero-relaxation verifier buys at most half a coverage point, because the ceiling is the certificate's own sublevel set, not the tightness of the bounding.

Result 2: an independent exact verifier ties the interval result. Everything above rests on one technology, interval arithmetic, so we built a second, exact one. It lifts a slice's two varying angles to $(\sin, \cos)$ pairs so the restriction of h is an exact polynomial of total degree five, then searches for an algebraic Positivstellensatz proof of $h < 0$ with no interval relaxation at all, implemented independently of the first verifier (`cvxpy`, Clarabel and SCS). On the identical polynomial it **ties interval branch-and-bound on every slice**, $93.6°/88.9°/85.6°$ against $93.8°/88.5°/85.6°$, mean coverage 22.0% against 22.0% (Figure 1e). An exact method landing where a conservative one does is evidence that the 93 to 106 degree band belongs to the certificate rather than to either verifier.

Result 3: radius and shape are separate questions. Coverage depends on two quantities, how far $h < 0$ can be proved and how well the certified region's geometry matches the true stable set at that radius. Coverage is defined through U's sublevel set alone, yet six gradient-based attempts to reshape U found nothing, because gradient descent was solving both questions inside one entangled non-convex objective. Decoupling them, by fixing the radius at each slice's own already-proved boundary and posing the shape question alone as a single convex program fitted to ground truth with a hard margin against every sampled unstable point, recovers an 86% relative gain over the classical shape, 39.8% against 21.4% at the same radius and with zero certified-unstable points (Figure 1f). Stated exactly, at every radius proved here the sampled ground truth inside each disk is entirely stable, since the nearest unstable point on slice $(5, 6)$ lies at $114.9°$, beyond any radius any method proved, so the fitted shape matches the naive disk almost exactly and the $+86\%$ measures how badly the classical shape under-uses a proved radius.

Transfer to a second, independently built topology. We built the IEEE/WSCC 9-bus three-generator system from MATPOWER's `case9.m`, with inertia and transient reactance cross-confirmed from two published sources, the power flow solved by `pandapower`, and the Kron reduction verified by an equilibrium-balance check (residual 10^{-16}) and a Hurwitz check. The physics baseline lands at 23.4% coverage and the identical learning procedure, reused unchanged, improves it to 32.3%, a 38% relative gain, both 100% sound (Figure 1c).

Negative results, reported as findings. Seven further independently designed attempts to extend the radius are in Table 1 and all failed. Their value is that they converge, since one exact verification technology and the most structurally permissive variation available in this family, a free $V(\boldsymbol{\phi}, \boldsymbol{\omega})$ verified over the full 18-D state with exact autograd $\dot{V}$, land on the same band as the fixed-structure baseline to one decimal place.

5 Discussion and limitations

The radius question is now well bounded because the failure modes are exhausted. Eight designs, one independent proof technology and one maximally permissive architecture all stop in the same 93 to 106 degree band. The shape question is open and is where sound gains still sit, since the convex program in Result 3 is a first minimal instance of a larger family, not yet tested at a radius where real boundary information exists to fit against.

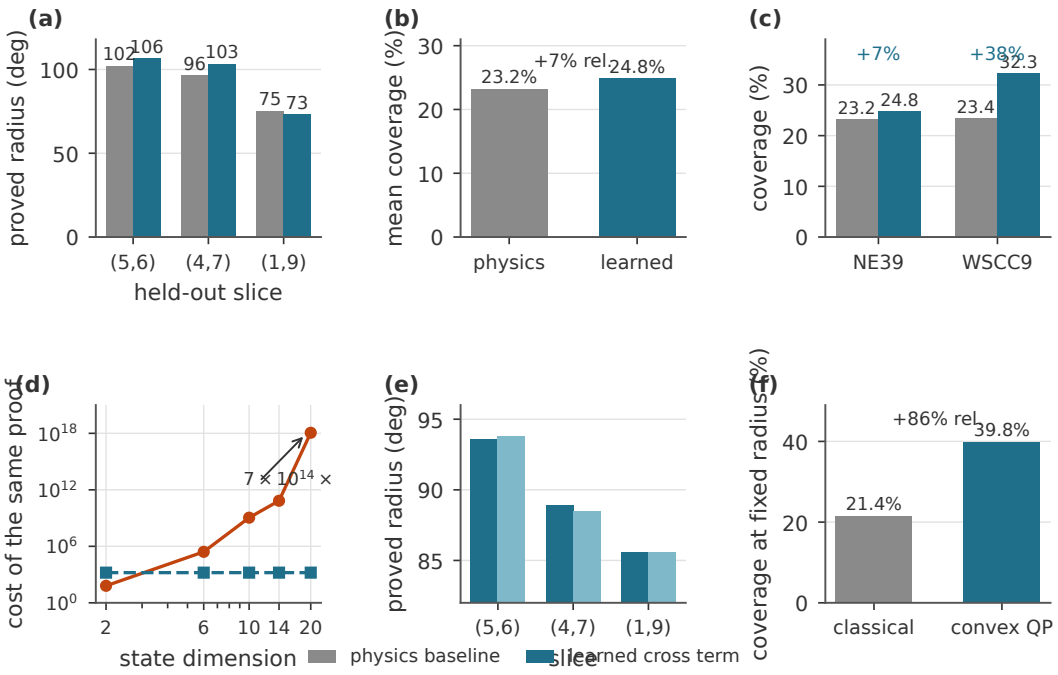


Figure 1: Six readouts from one protocol, all 100% sound. Held-out proved radius per slice (a) and mean coverage (b) for the physics baseline against the learned cross term, $n = 5{,}041$ Monte Carlo points per slice with $4{,}448/4{,}154/4{,}081$ stable; the same comparison on a separately rebuilt three-generator topology (c), where the procedure gains 38% relative against 7% on NE39; measured monolithic branch-and-bound leaf-box cost against compositional operation count for one claim, 2-D to 20-D, log-log (d); the exact sum-of-squares certificate against interval branch-and-bound on the identical slice polynomial, tied within 0.4° (e); and coverage at each slice's own fixed proved radius, classical transient-energy shape against the convex fit (f).

Table 1: The experiment ledger behind the claim that the radius is bounded by the certificate rather than by the verifier. Coverage is 100% sound throughout; the last two rows are the positive results.

ID	Tested	Coverage
c8	Scalar potential residual (gradient-only lever)	9.4%
c14	Exact closed-form cross-term cancellation, zero parameters	0.1%
c15	Potential reshaping, frozen and jointly trained cross term	16.8%
c16	Full-state visibility, double capacity, no locality	17.4%
c17	Free $V(\phi, \omega)$ over the full 18-D state	15.7%
c18	Independent exact SOS verifier, all three slices	22.0%
c19	Convex shape search at each slice's proved radius	39.8%

Limitations. Coverage is a 2-D-slice metric, not a volume estimate of the 9-D or 18-D state space, so it does not by itself license a claim about the full power system. The central neighbourhood is not verified by branch-and-bound and rests on the local-Hessian-plus-sampling argument, the single sampled ingredient in the paper. The shape containment check is sampling-based and shows a borderline 0.006 rad leak on one slice, smaller than that grid's own spacing, and this is the one non-rigorous step in the study. Under a $\pm 15\%$ loading spread with no retraining the trained field moves by under 5% on two of three slices and collapses from 72.9° to 0° on the third, so the transfer result transfers the procedure rather than guaranteeing local robustness. The plant is Kron-reduced at a fixed equilibrium, so nothing here speaks to switching topology, and branch-and-bound soundness is modulo floating point.

Broader impact. The quantity deciding how much renewable generation a network absorbs is the size of the set of post-fault states its controller returns to synchronism, and composition is what makes that set provable at ten generators instead of two, which is the boundary this study measures.

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Availability of code and data

Every number reported here is read from the committed per-run result records by a single checking script that also asserts the headline values it plots, so a stale figure fails loudly rather than shipping. Code, records and the second topology's build script are available from the author on request. The plant reuses the public benchmark of (Cui and Zhang, 2022) verbatim and the second topology is rebuilt from public sources, so no non-public data is used.

Use of AI tools

Generative AI tools were used for review and polish rather than for drafting: to audit the manuscript, to re-check numerical claims by re-execution against the saved records rather than by reading, and to confirm that the second verifier was implemented independently of the first so that the tie in Result 2 is a cross-check rather than a refactor. The authors directed the work and take responsibility for all content.